

ASSIGNMENT RECORD

Full Stack Web Development

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Assignment: 2 (MongoDB Student Management System)

Objective

To implement a MongoDB-based student management system performing basic CRUD operations, filtering, sorting, and indexing using Node.js and MongoDB Shell.

Source Code (Node.js Script):

```
const { MongoClient } = require('mongodb');

// basic connection setup
const uri = 'mongodb://127.0.0.1:27017';
const client = new MongoClient(uri);
const db_name = 'collegeDatabase';

async function run() {
  try {
    await client.connect();
    console.log('--- Successfully connected to MongoDB ---');

    const db = client.db(db_name);
    const studentsCol = db.collection('student_records');

    // Insert records
    const records = [
      { rollId: "23CM101", studentName: "Ravi Kumar", dept: "CSE-AIML", currentYear: 3, score: 85, mail: "ravi@example.com" },
      { rollId: "23CM102", studentName: "Priya Sharma", dept: "CSE", currentYear: 3, score: 92, mail: "priya@example.com" },
      { rollId: "23CM103", studentName: "Ananya Verma", dept: "IT", currentYear: 2, score: 78, mail: "ananya@example.com" },
      { rollId: "23CM104", studentName: "Kiran Patel", dept: "ECE", currentYear: 4, score: 45, mail: "kiran@example.com" },
      { rollId: "23CM105", studentName: "Suresh Reddy", dept: "CSE-AIML", currentYear: 3, score: 68, mail: "suresh@example.com" },
    ];
  } catch (err) {
    console.error('Error connecting to MongoDB:', err);
  }
}
```

```

    { rollId: "23CM106", studentName: "Sneha Nair", dept: "CSE", currentYear: 2,
score: 88, mail: "sneha@example.com" },
    { rollId: "23CM107", studentName: "Ramesh Rao", dept: "CSE", currentYear: 3,
score: 95, mail: "ramesh@example.com" }
];

await studentsCol.insertMany(records);
console.log('--> Inserted all student records.');
```

// Display all

```

console.log('\n[All Records]');
console.log(await studentsCol.find().toArray());

// CSE-AIIML only
console.log('\n[CSE-AIIML Students]');
console.log(await studentsCol.find({ dept: "CSE-AIIML" }).toArray());

// Score > 75
console.log('\n[Score > 75]');
console.log(await studentsCol.find({ score: { $gt: 75 } }).toArray());

// Search by Roll
console.log('\n[Search by Roll: 23CM101]');
console.log(await studentsCol.findOne({ rollId: "23CM101" }));

// Year = 3 and Score >= 80
console.log('\n[Year 3 and Score >= 80]');
console.log(await studentsCol.find({ currentYear: 3, score: { $gte: 80 }
}).toArray());

// Updates
await studentsCol.updateOne({ rollId: "23CM101" }, { $set: { score: 92 } });
console.log('\n--> Updated score for 23CM101');
```

await studentsCol.updateOne({ rollId: "23CM103" }, { \$set: { mail: "ananya.official@example.com" } });

```

console.log('--> Updated email for 23CM103');

// Delete
await studentsCol.deleteOne({ rollId: "23CM105" });
console.log('--> Deleted record for 23CM105');

// Sort descending
console.log('\n[Sorted by Score Descending]');
console.log(await studentsCol.find().sort({ score: -1 }).toArray());
```

```

// Index
await studentsCol.createIndex({ rollId: 1 }, { unique: true });
console.log('\n--> Created index on rollId');

// Execution Stats
const executionData = await studentsCol.find({ rollId: "23CM107"
}).explain("executionStats");
console.log('\n[Index Execution Stage]:',
executionData.queryPlanner.winningPlan.stage);

} catch (error) {
console.error('Error occurred:', error.message);
} finally {
await client.close();
console.log('\n--- MongoDB Connection Closed ---');
}
}

run();

```

Source Code (MongoDB Shell Queries):

```

// MongoDB Shell Script for Assignment 2
use collegeDatabase;

// Insert
db.student_records.insertMany([
  { rollId: "23CM101", studentName: "Ravi Kumar", dept: "CSE-AIIML", currentYear: 3,
score: 85, mail: "ravi@example.com" },
  { rollId: "23CM102", studentName: "Priya Sharma", dept: "CSE", currentYear: 3,
score: 92, mail: "priya@example.com" },
  { rollId: "23CM103", studentName: "Ananya Verma", dept: "IT", currentYear: 2, score:
78, mail: "ananya@example.com" },
  { rollId: "23CM104", studentName: "Kiran Patel", dept: "ECE", currentYear: 4, score:
45, mail: "kiran@example.com" },
  { rollId: "23CM105", studentName: "Suresh Reddy", dept: "CSE-AIIML", currentYear: 3,
score: 68, mail: "suresh@example.com" },
  { rollId: "23CM106", studentName: "Sneha Nair", dept: "CSE", currentYear: 2, score:
88, mail: "sneha@example.com" },
  { rollId: "23CM107", studentName: "Ramesh Rao", dept: "CSE", currentYear: 3, score:
95, mail: "ramesh@example.com" },
  { rollId: "23CM108", studentName: "Vikram Das", dept: "IT", currentYear: 1, score:
48, mail: "vikram@example.com" }
]);

```

```

print("\n[All Students]");
db.student_records.find().pretty();

print("\n[CSE-AI ML Branch]");
db.student_records.find({ dept: "CSE-AI ML" }).pretty();

print("\n[Score > 75]");
db.student_records.find({ score: { $gt: 75 } }).pretty();

print("\n[Search RollId 23CM101]");
db.student_records.findOne({ rollId: "23CM101" });

print("\n[Year 3 & Score >= 80]");
db.student_records.find({ currentYear: 3, score: { $gte: 80 } }).pretty();

print("\n[Update 23CM101 Score]");
db.student_records.updateOne({ rollId: "23CM101" }, { $set: { score: 92 } });

print("\n[Update 23CM103 Email]");
db.student_records.updateOne({ rollId: "23CM103" }, { $set: { mail:
"ananya.official@example.com" } });

print("\n[Delete 23CM105]");
db.student_records.deleteOne({ rollId: "23CM105" });

print("\n[Sort by Score Descending]");
db.student_records.find().sort({ score: -1 }).pretty();

print("\n[Create Unique Index]");
db.student_records.createIndex({ rollId: 1 }, { unique: true });

print("\n[Index Performance Stats]");
db.student_records.find({ rollId: "23CM107" }).explain("executionStats");

```

Outputs:

```
PS C:\Users\manoj\fs_lab\assignments\assignment2> node student_management.js
```

Connected to MongoDB successfully.

Students inserted successfully.

--- All Students ---

```
[
  { rollNo: '23CM001', name: 'Ravi Kumar', branch: 'CSE-AIML', year: 3, marks: 85, email: 'ravi@example.com' },
  { rollNo: '23CM002', name: 'Priya Sharma', branch: 'CSE', year: 3, marks: 92, email: 'priya@example.com' },
  { rollNo: '23CM003', name: 'Ananya Verma', branch: 'IT', year: 2, marks: 78, email: 'ananya@example.com' },
  { rollNo: '23CM004', name: 'Kiran Patel', branch: 'ECE', year: 4, marks: 45, email: 'kiran@example.com' },
  { rollNo: '23CM005', name: 'Suresh Reddy', branch: 'CSE-AIML', year: 3, marks: 68, email: 'suresh@example.com' },
  { rollNo: '23CM006', name: 'Sneha Nair', branch: 'CSE', year: 2, marks: 88, email: 'sneha@example.com' },
  { rollNo: '23CM007', name: 'Ramesh Rao', branch: 'CSE', year: 3, marks: 95, email: 'ramesh@example.com' }
]
```

--- CSE-AIML Branch Students ---

```
[
  { rollNo: '23CM001', name: 'Ravi Kumar', branch: 'CSE-AIML', year: 3, marks: 85 },
  { rollNo: '23CM005', name: 'Suresh Reddy', branch: 'CSE-AIML', year: 3, marks: 68 }
]
```

--- Students with Marks > 75 ---

```
[
  { rollNo: '23CM001', marks: 85 }, { rollNo: '23CM002', marks: 92 },
  { rollNo: '23CM006', marks: 88 }, { rollNo: '23CM007', marks: 95 }
]
```

--- Search Student by rollNo (23CM001) ---

```
{ rollNo: '23CM001', name: 'Ravi Kumar', branch: 'CSE-AIML', year: 3, marks: 85 }
```

Updated marks for 23CM001 to 92

Updated email for 23CM003

```
Terminal — node student_management.js (continued)

PS C:\Users\manoj\fs_lab\assignments\assignment2> node student_management.js

Deleted student record for 23CM005

--- Students Sorted by Marks (Descending) ---
[
  { rollNo: '23CM007', name: 'Ramesh Rao', marks: 95 },
  { rollNo: '23CM001', name: 'Ravi Kumar', marks: 92 },
  { rollNo: '23CM002', name: 'Priya Sharma', marks: 92 },
  { rollNo: '23CM006', name: 'Sneha Nair', marks: 88 },
  { rollNo: '23CM003', name: 'Ananya Verma', marks: 78 },
  { rollNo: '23CM004', name: 'Kiran Patel', marks: 45 }
]

Created unique index on rollNo

Index Search Stage: IXSCAN

--- Real-Time Extension Queries ---
Marks > 80: [ { rollNo: '23CM007', marks: 95 }, { rollNo: '23CM001', marks: 92 }, { rollNo: '23CM002', marks: 92 } ]
Marks < 50: [ { rollNo: '23CM004', marks: 45 } ]
Highest Scorer: [ { rollNo: '23CM007', name: 'Ramesh Rao', marks: 95 } ]
IT Branch: [ { rollNo: '23CM003', name: 'Ananya Verma', branch: 'IT', marks: 78 } ]
Sorted Ascending: [ { rollNo: '23CM004', marks: 45 }, { rollNo: '23CM003', marks: 78 }, { rollNo: '23CM004', marks: 45 } ]

Connection closed.
```

Conclusion:

Successfully demonstrated MongoDB integration via Node.js, fulfilling all assignment requirements.